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Ex No – 9	Deadlock Avoidance
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Program :

```
#include <stdio.h>

int main() {

    // P0, P1, P2, P3, P4 are the Process names here

    int n, m, i, j, k;

    n = 5; // Number of processes
    m = 3; // Number of resources

    int alloc[5][3] = {

        {0, 1, 0}, // P0 // Allocation Matrix

        {2, 0, 0}, // P1

        {3, 0, 2}, // P2

        {2, 1, 1}, // P3

        {0, 0, 2} // P4

    };

    int max[5][3] = {

        {7, 5, 3}, // P0 // MAX Matrix

        {3, 2, 2}, // P1

        {9, 0, 2}, // P2

        {2, 2, 2}, // P3

        {4, 3, 3} // P4

    };

    int avail[3] = {3, 3, 2}; // Available Resources

    int f[n], ans[n], ind = 0;
```

```

for (k = 0; k < n; k++) {
    f[k] = 0;
}
int need[n][m];
for (i = 0; i < n; i++) {
    for (j = 0; j < m; j++) {
        need[i][j] = max[i][j] - alloc[i][j];
    }
}
int y = 0;
for (k = 0; k < 5; k++) {
    for (i = 0; i < n; i++) {
        if (f[i] == 0) {
            int flag = 0;
            for (j = 0; j < m; j++) {
                if (need[i][j] > avail[j]) {
                    flag = 1;
                    break;
                }
            }
            if (flag == 0) {
                ans[ind++] = i;
                for (y = 0; y < m; y++) {
                    avail[y] += alloc[i][y];
                }
                f[i] = 1;
            }
        }
    }
}

```

```
    }  
    }  
}  
printf("The SAFE Sequence is:\n");  
for (i = 0; i < n - 1; i++) {  
    printf(" P%d ->", ans[i]);  
}  
printf(" P%d\n", ans[n - 1]);  
return 0;  
}
```

Output :

```
The SAFE Sequence is:  
P1 -> P3 -> P4 -> P0 -> P2  
[cse32@localhost ~]$
```